

Company Policies and Meme Inflation

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1. Introduction

Nowadays every consumer product is sold and consumed faster compared to the past. People buy new clothes more frequently, opt for more rapid food options (Didarloo et al., 2022) and the list goes on.

This project investigates how this faster consuming behavior reflects on cultural elements, specifically the internet memes in this case. We will examine the relation between the introduction of vertical short video user interfaces and trend lifespan of memes i.e. how many weeks a meme stays on trends. Instagram announced its vertical short video user interface “Reels” on August 5th, 2020 (Instagram, 2020). Shortly after, YouTube released its own version called “Shorts” on September 14th same year (YouTube, 2020). Additionally, the date of acquisition of Musical.ly by ByteDance in order to merge it with TikTok which also has vertical short-video format in 2017 is also close to others (Reuters, 2017).

We will claim that memes that emerged after 2020 lasts shorter on trends compared to the memes that emerged before 2020 as the most prominent social media apps released their vertical video UI around 2020. Hence:

Null Hypothesis, H_0 : There is no difference between the trend lifespan of the pre-2020 memes and that of post-2020 memes

Alternative Hypothesis, H_A : Trend lifespan of post-2020 memes are shorter than that of pre-2020 memes

2. Data Collection

KnowYourMeme is a website where voluntary staff can upload the information related to the internet memes. The website has the origin year, origin platform, and other detailed information related to 16975 memes in total. BeautifulSoup and request were used to fetch the name of every 25th meme starting from the oldest one. This has created a list of 1785 names, which was used to create special URLs to the pages of each meme. Finally, origin_year and origin_platform were fetched from these pages.

Later, Google Trends' python library "pytrends" was utilized to get the data related to the search interest for 5 years from a meme's first appearance. A meme was considered on trends when search interest is more than 15% of its maximum interests. At the end of the process, a csv file including "name", "origin_year", "origin_platform" and "weeks_of_trends" was obtained.

3. Data Cleaning and Data Analysis

When the data is inspected one could see that there are memes that were on trends for 0 and 235 weeks, which are the shortest and longest values in the dataset respectively. The reason for 0 weeks is that some memes became popular much later after they emerge (e.g. a meme from a 1984 movie is considered to be on trends for 0 weeks). 235 trending weeks occurred due to the memes that are uploaded on the website with a name similar to a thing which is a general public interest over longer years (e.g. Drake). Therefore, the rows including these two were cleaned before starting the analysis.

First of all, the most evident thing is that newer memes emerge from different platforms compared to the past. Figure 1 shows that YouTube has lost its leadership to Twitter in terms of being home to the newer memes. Additionally, some platforms such as 4chan become obscure after 2020.

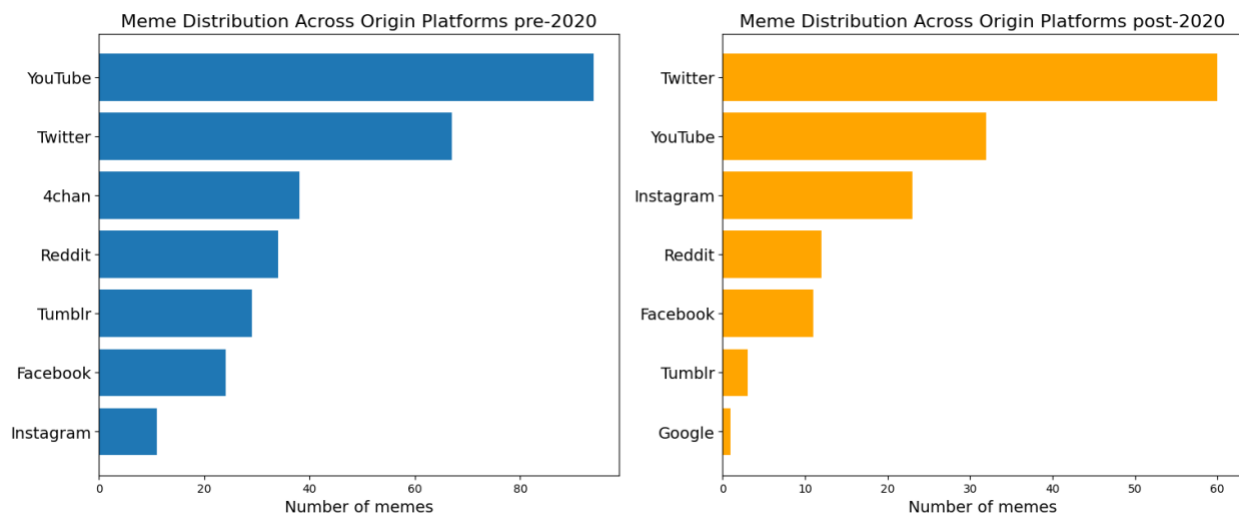


Figure 1. Meme Distribution Across Platforms

Best performing platform includes Newgrounds and Vine within the context of having the memes with longest lifespan on trends (Figure 2). Nevertheless, they

are not among the platforms that are actively used today as it can be seen from Figure 1.

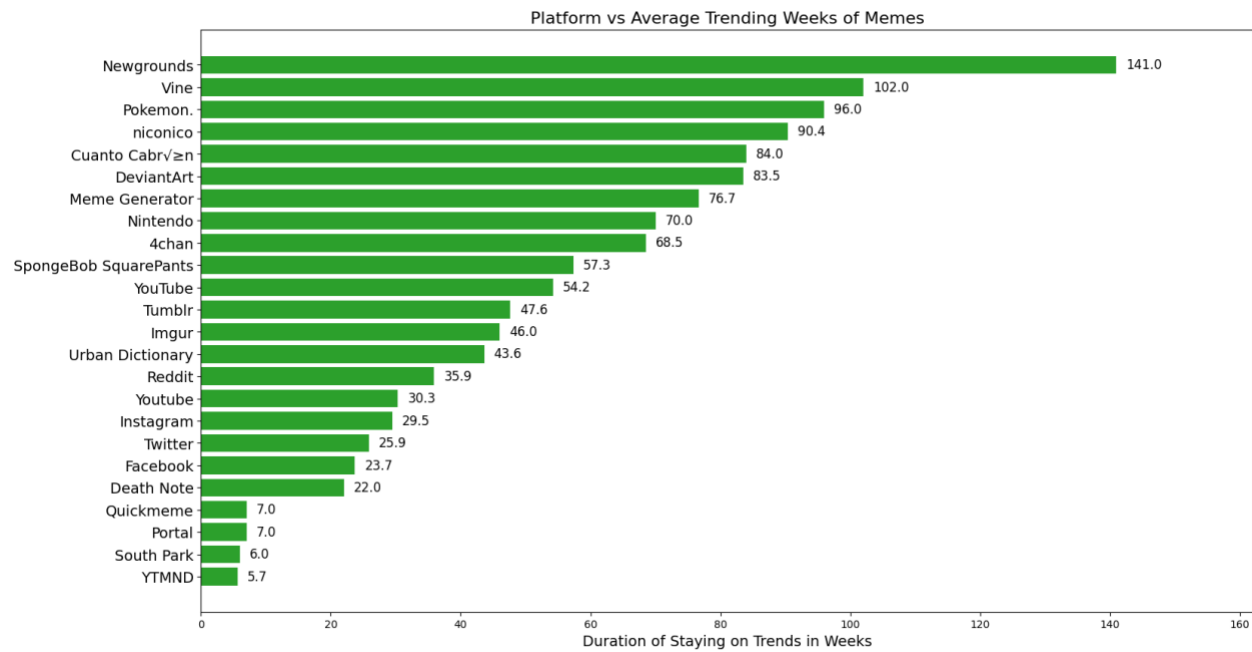


Figure 2. Platforms performing best in terms of meme lifespan on trends

Number of memes from pre-2020 and post-2020 has an unfortunate property. As we are still in 2025, this year seems to have the least number of memes compared to others. Figure 3 demonstrates a peak in 2019-2020, which coincides with the Covid-19 lockdown.

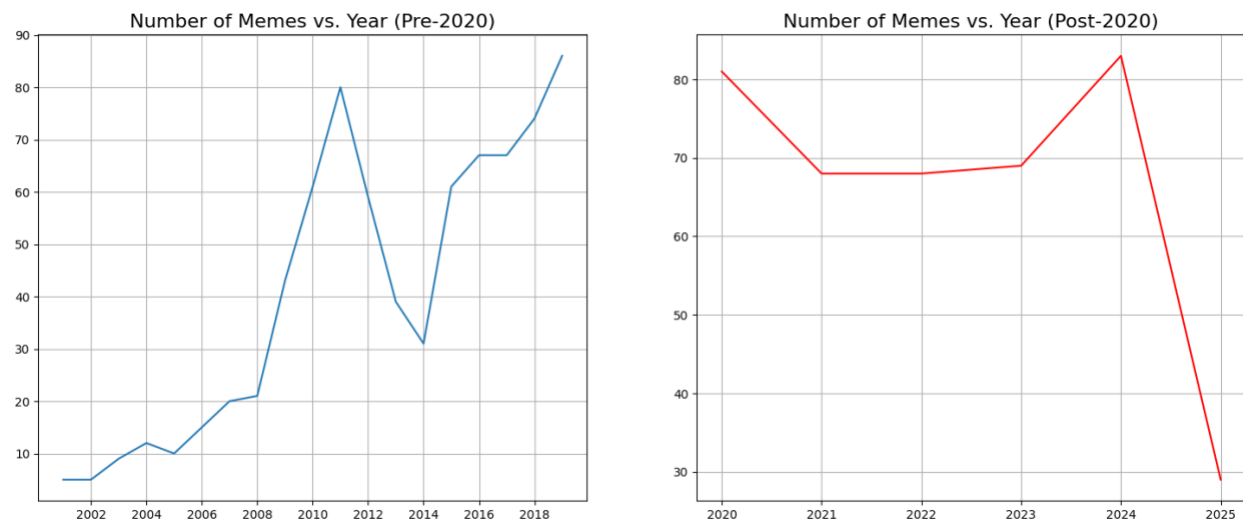


Figure 3. Number of Memes from 2000-2025

4. Hypothesis Testing

We need to compare the means of the number of trending weeks of pre-2020 and post-2020 memes. Figure 4 indicates that even though the medians of two group is close to each other, there is a drastic difference between how they distribute.

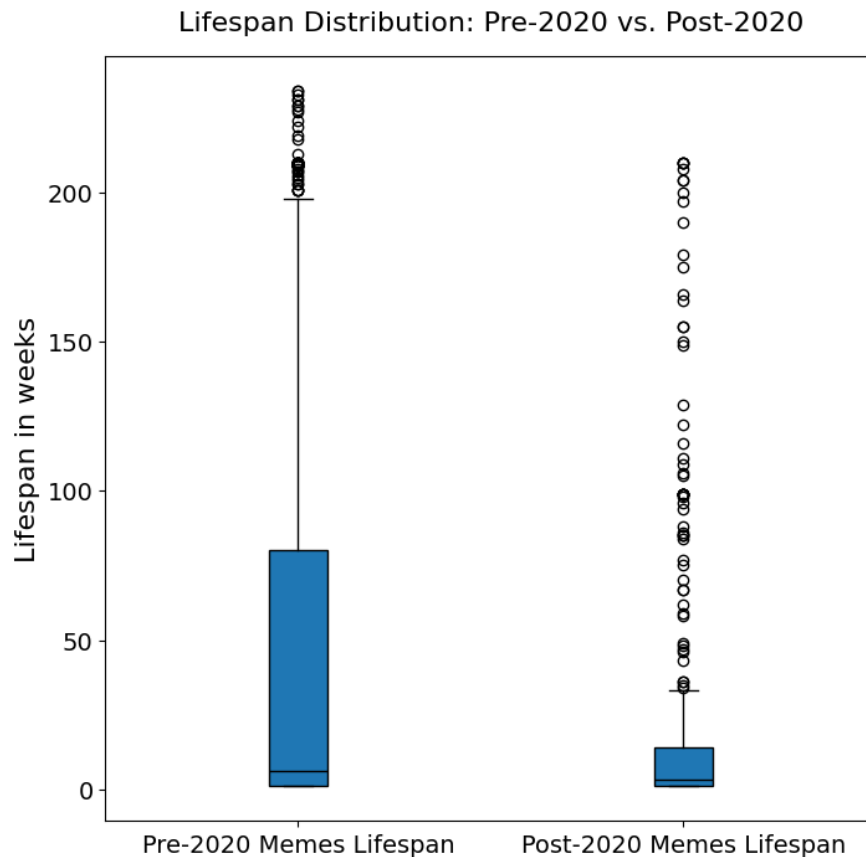


Figure 4. Lifespan of pre and post- 2020 memes

In order to find out whether there is a significant difference between the means of two groups, t-test or Mann-Whitney U test should be applied. Since distributions are not normal, we can not use the t-test.

Mann-Whitney U test that checks whether the pre-2020 memes trend lifespan is greater than that of post-2020 memes yielded a **P-value of $10E-7$** which is smaller than significance value of 0.05. **Hence, we can reject the null hypothesis.** This means that newer memes last shorter on trends.

5. Machine Learning

The dataset created for this project contains only 4 columns, and among them, “weeks_of_trends” is the only target that is suitable for regression and prediction. We cannot apply any classification models as any of the non-numerical features do not have a significant relation with other features. So as to decide which regression model to harness, we have observed the behavior between “weeks-of-trends” and “origin_year” (Figure 5).

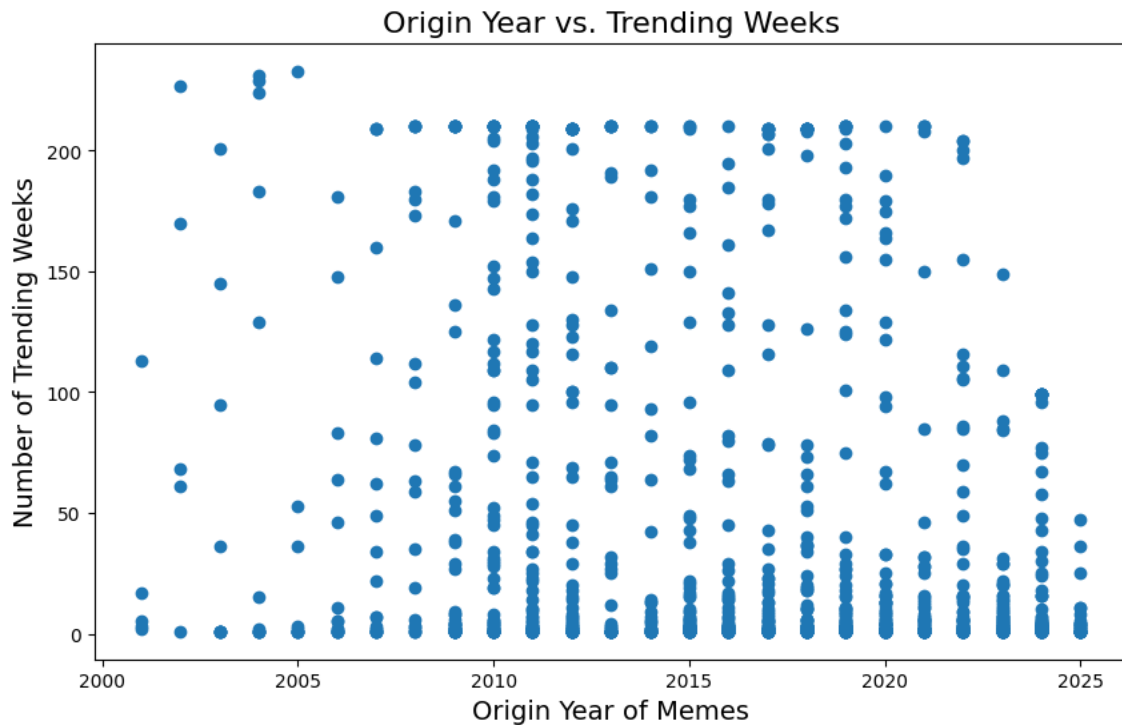


Figure 5. Trending Weeks of Memes from Every Year Between 2000-2025

There is no linear trend, and the data is highly disordered. Hence, we can use Random Forest Regressor and Extreme Gradient Boosting (XGBoost) Regressor. First, label encoding was applied to “origin_platform” in order to make this feature numerical because the models are not able to comprehend verbal expressions. Then, the dataset was split into 80% training split and 20% test split in random.

To find the best hyperparameters for Random Forest, number of estimators between 90-120 and maximum depth between 1-10 were tested in a loop, and the combination with highest R2 (so lowest MSE) was selected. **This model yielded an R2 of 0.057**, which cannot be considered promising at all.

Finally, XGBoost which handles empty values better naturally was employed. Since “origin_platform” feature had empty values the expected R2 value was higher for this model. Learning rate was optimized through iterations of different values ranging from 0.1-1 (Figure 6). However, **the best R2 value was 0.007**, which is worse than Random Forest.

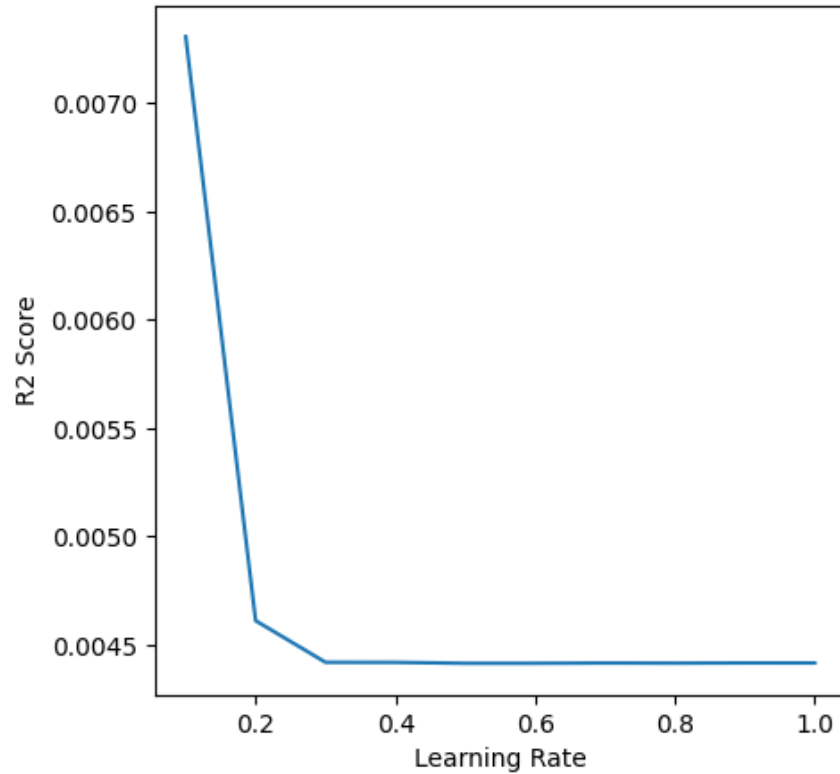


Figure 6. R2 with different Learning Rates of XGBoost

6. Challenges and Discussion for the Future Works

Two main problems were encountered in this project. First one is that some memes emerged after 2020 did not have enough time to show their potential. For example, a meme from 2025 can only have about 40 weeks on trends at maximum. Probably these instances are the reason why the p-value is drastically smaller than the significance value.

Other challenge is that the dataset did not contain enough columns to let machine learning models to understand the underlying relation between the features. Even though the dataset included information related to 1785 memes, the “origin_platform” and “origin_year” features were not enough to explain this decrease in number of trending weeks.

7. References

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